THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 84	Area $= l \times b$.
2	**(b)** 46	Perimeter $= 2(l + b)$.
3	**(c)** 36	Perimeter $= 4\times$ side.
4	**(d)** 48	Area $= \frac{1}{2}\times$ base $\times$ height.
5	**(a)** 120	Area $=$ base $\times$ height.
6	**(a)** 44	$C = 2\pi r$.
7	**(a)** 154	$A = \pi r^2$.
8	**(d)** 7	$r = \frac{C}{2\pi}$.
9	**(d)** 135	Area $= l \times b$.
10	**(c)** 62	Perimeter $= 2(l + b)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$C = 2\pi r = 2 \times \frac{22}{7} \times 21$ (1); $C = 132$ cm (1).	1+1
12	$A = \pi r^2 = \frac{22}{7} \times 7 \times 7$ (1); $A = 154$ cm^2 (1).	1+1
13	$A = \frac{1}{2} \times 18 \times 10$ (1); $A = 90$ cm^2 (1).	1+1
14	$C = 2\pi r = 2 \times \frac{22}{7} \times 42$ (1); $C = 264$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $=$ base $\times$ height (1); $108 = 12 \times h$ (1); $h = 9$ cm (1).	1+1+1
16	$s = \frac{3+4+5}{2} = 6$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 6$ cm^2 (1).	1+1+1
17	$A = \frac{1}{2}(a+b)h$ (1); $A = \frac{1}{2} \times (10 + 14) \times 6$ (1); $A = 72$ cm^2 (1).	1+1+1
	OR: Area $=$ base $\times$ height (1); $120 = 15 \times h$ (1); $h = 8$ cm (1).	1+1+1
18	$s = \frac{5+12+13}{2} = 15$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 30$ cm^2 (1).	1+1+1
	OR: $A = \frac{1}{2}(a+b)h$ (1); $A = \frac{1}{2} \times (12 + 18) \times 8$ (1); $A = 120$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Perimeter $= 2(40 + 25) = 130$ m (1); Area $= 40 \times 25 = 1000$ m^2 (1); Cost $= 130 \times 12$ (1); $=$ ₹1560 (1); Units stated (1).	1+1+1+1+1
	OR: $C = 2\pi r = 88$ m (1); $A = \pi r^2 = 616$ m^2 (1); Cost $= 88 \times 15$ (1); $=$ ₹1320 (1); Units stated (1).	1+1+1+1+1
20	Perimeter $= 2(60 + 35) = 190$ m (1); Area $= 60 \times 35 = 2100$ m^2 (1); Cost $= 190 \times 15$ (1); $=$ ₹2850 (1); Units stated (1).	1+1+1+1+1
	OR: $C = 2\pi r = 132$ m (1); $A = \pi r^2 = 1386$ m^2 (1); Cost $= 132 \times 12$ (1); $=$ ₹1584 (1); Units stated (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up circumference ($2\pi r$) and area (πr^2), or writing the area in cm instead of cm^2.
- In Heron's formula, using the perimeter instead of the semi-perimeter s in the factors $(s - a)$.
- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 52	Perimeter $= 4\times$ side.
2	**(d)** 72	Area $= \frac{1}{2}\times$ base $\times$ height.
3	**(c)** 156	Area $=$ base $\times$ height.
4	**(a)** 88	$C = 2\pi r$.
5	**(c)** 616	$A = \pi r^2$.
6	**(a)** 21	$r = \frac{C}{2\pi}$.
7	**(b)** 108	Area $= l \times b$.
8	**(d)** 58	Perimeter $= 2(l + b)$.
9	**(b)** 68	Perimeter $= 4\times$ side.
10	**(c)** 77	Area $= \frac{1}{2}\times$ base $\times$ height.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$A = \pi r^2 = \frac{22}{7} \times 14 \times 14$ (1); $A = 616$ cm^2 (1).	1+1
12	$A = \frac{1}{2} \times 24 \times 7$ (1); $A = 84$ cm^2 (1).	1+1
13	$C = 2\pi r = 2 \times \frac{22}{7} \times 56$ (1); $C = 352$ cm (1).	1+1
14	$A = \pi r^2 = \frac{22}{7} \times 35 \times 35$ (1); $A = 3850$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $=$ base $\times$ height (1); $140 = 20 \times h$ (1); $h = 7$ cm (1).	1+1+1
16	$s = \frac{6+8+10}{2} = 12$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 24$ cm^2 (1).	1+1+1
17	$A = \frac{1}{2}(a+b)h$ (1); $A = \frac{1}{2} \times (9+15) \times 10$ (1); $A = 120$ cm^2 (1).	1+1+1
	OR: Area $=$ base $\times$ height (1); $176 = 16 \times h$ (1); $h = 11$ cm (1).	1+1+1
18	$s = \frac{9+12+15}{2} = 18$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 54$ cm^2 (1).	1+1+1
	OR: $A = \frac{1}{2}(a+b)h$ (1); $A = \frac{1}{2} \times (16+20) \times 5$ (1); $A = 90$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Perimeter $= 2(45+30) = 150$ m (1); Area $= 45 \times 30 = 1350$ m^2 (1); Cost $= 150 \times 20$ (1); $=$ ₹3000 (1); Units stated (1).	1+1+1+1+1
	OR: $C = 2\pi r = 176$ m (1); $A = \pi r^2 = 2464$ m^2 (1); Cost $= 176 \times 10$ (1); $=$ ₹1760 (1); Units stated (1).	1+1+1+1+1
20	Perimeter $= 2(50+28) = 156$ m (1); Area $= 50 \times 28 = 1400$ m^2 (1); Cost $= 156 \times 18$ (1); $=$ ₹2808 (1); Units stated (1).	1+1+1+1+1
	OR: $C = 2\pi r = 220$ m (1); $A = \pi r^2 = 3850$ m^2 (1); Cost $= 220 \times 8$ (1); $=$ ₹1760 (1); Units stated (1).	1+1+1+1+1

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- Using a slanting side instead of the perpendicular height in the area of a parallelogram or triangle.
- Treating the width of a path as the outer radius; the outer radius is $r + w$ and the area of the path is the *difference* of the two circle areas.
- Forgetting the factor $\frac{\theta}{360}$ in the arc length formula, or using the area formula πr^2 for arc length.
- Applying Brahmagupta's formula to a quadrilateral that is not cyclic.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 126	Area = base × height.
2	**(b)** 132	$C = 2\pi r$.
3	**(b)** 1386	$A = \pi r^2$.
4	**(b)** 35	$r = \frac{C}{2\pi}$.
5	**(b)** 200	Area $= l \times b$.
6	**(a)** 84	Perimeter $= 2(l + b)$.
7	**(b)** 100	Perimeter $= 4\times$ side.
8	**(b)** 70	Area $= \frac{1}{2}\times$ base × height.
9	**(c)** 100	Area = base × height.
10	**(d)** 220	$C = 2\pi r$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$A = \frac{1}{2} \times 30 \times 16$ (1); $A = 240$ cm^2 (1).	1+1
12	$C = 2\pi r = 2 \times \frac{22}{7} \times 63$ (1); $C = 396$ cm (1).	1+1
13	$A = \pi r^2 = \frac{22}{7} \times 42 \times 42$ (1); $A = 5544$ cm^2 (1).	1+1
14	$A = \frac{1}{2} \times 22 \times 9$ (1); $A = 99$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area = base × height (1); $150 = 25 \times h$ (1); $h = 6$ cm (1).	1+1+1
16	$s = \frac{8+15+17}{2} = 20$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 60$ cm^2 (1).	1+1+1
17	$A = \frac{1}{2}(a+b)h$ (1); $A = \frac{1}{2} \times (7+13) \times 12$ (1); $A = 120$ cm^2 (1).	1+1+1
	OR: Area = base × height (1); $234 = 18 \times h$ (1); $h = 13$ cm (1).	1+1+1
18	$s = \frac{7+24+25}{2} = 28$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 84$ cm^2 (1).	1+1+1
	OR: $A = \frac{1}{2}(a+b)h$ (1); $A = \frac{1}{2} \times (11+19) \times 4$ (1); $A = 60$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Perimeter $= 2(72+48) = 240$ m (1); Area $= 72 \times 48 = 3456$ m^2 (1); Cost $= 240 \times 10$ (1); $=$ ₹2400 (1); Units stated (1).	1+1+1+1+1
	OR: $C = 2\pi r = 264$ m (1); $A = \pi r^2 = 5544$ m^2 (1); Cost $= 264 \times 5$ (1); $=$ ₹1320 (1); Units stated (1).	1+1+1+1+1
20	Perimeter $= 2(55+45) = 200$ m (1); Area $= 55 \times 45 = 2475$ m^2 (1); Cost $= 200 \times 25$ (1); $=$ ₹5000 (1); Units stated (1).	1+1+1+1+1
	OR: $C = 2\pi r = 308$ m (1); $A = \pi r^2 = 7546$ m^2 (1); Cost $= 308 \times 6$ (1); $=$ ₹1848 (1); Units stated (1).	1+1+1+1+1

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